KAIZEN MATHS

Lesson starter

Advanced Differentiation

Parametric Differentiation

Name _____

S1

For $x(t) = 3t^2 - 3t - 3$ and $y(t) = 2t^3 + 4t^2 - t + 2$, find $\frac{dy}{dx}$.

S2

For $x(t) = 3t^2 - 2t$ and $y(t) = -4t^3 - 2t^2$, find $\frac{d^2y}{dx^2}$.

S3

Find tangent and normal at $t = 1$ for $x(t) = 4t^2 + 1$ and $y(t) = -3t^3 + 4t + 1$.

S4

For $x(t) = -2t^2 + 3t$ and $y(t) = -4t^3 - 2t$, find where tangents are horizontal or vertical.

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Find tangent and normal at $t = 1$ for $x(t) = 4t^2 + 1$ and $y(t) = -3t^3 + 4t + 1$.

S4

For $x(t) = -2t^2 + 3t$ and $y(t) = -4t^3 - 2t$, find where tangents are horizontal or vertical.

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KAIZEN MATHS

Exit ticket

Advanced Differentiation

Parametric Differentiation

Name _____

E1

Express the area from $t = 1$ to $t = 4$ for $x(t) = -4t^2, y(t) = -2t - 4$.

E2

For $x(t) = -2t^2 - 4t$ and $y(t) = -3t^3 - 2t^2 + 4t + 2$, find $\frac{dy}{dx}$.

E3

For $x(t) = -2t^2$ and $y(t) = -3t^3 - 3t^2$, find $\frac{d^2y}{dx^2}$.

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Parametric Differentiation

Name _____

E1

Express the area from $t = 1$ to $t = 4$ for $x(t) = -4t^2, y(t) = -2t - 4$.

E2

For $x(t) = -2t^2 - 4t$ and $y(t) = -3t^3 - 2t^2 + 4t + 2$, find $\frac{dy}{dx}$.

E3

For $x(t) = -2t^2$ and $y(t) = -3t^3 - 3t^2$, find $\frac{d^2y}{dx^2}$.

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KAIZEN MATHS

Exit ticket

Advanced Differentiation

Parametric Differentiation

Name _____

E1

Express the area from $t = 1$ to $t = 4$ for $x(t) = -4t^2, y(t) = -2t - 4$.

E2

For $x(t) = -2t^2 - 4t$ and $y(t) = -3t^3 - 2t^2 + 4t + 2$, find $\frac{dy}{dx}$.

E3

For $x(t) = -2t^2$ and $y(t) = -3t^3 - 3t^2$, find $\frac{d^2y}{dx^2}$.

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KAIZEN MATHS

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Advanced Differentiation

Parametric Differentiation

Name _____

E1

Express the area from $t = 1$ to $t = 4$ for $x(t) = -4t^2, y(t) = -2t - 4$.

E2

For $x(t) = -2t^2 - 4t$ and $y(t) = -3t^3 - 2t^2 + 4t + 2$, find $\frac{dy}{dx}$.

E3

For $x(t) = -2t^2$ and $y(t) = -3t^3 - 3t^2$, find $\frac{d^2y}{dx^2}$.

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Teacher answers & working

Advanced Differentiation · Parametric Differentiation

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S1

For $x(t) = 3t^2 - 3t - 3$ and $y(t) = 2t^3 + 4t^2 - t + 2$, find $\frac{dy}{dx}$.

Answer:

$$\frac{dy}{dx} = \frac{6t^2 + 8t - 1}{6t - 3}$$

Working:

For parametric equations, differentiate both x and y with respect to t.

$$\frac{dx}{dt} = 6t - 3$$

$$\frac{dy}{dt} = 6t^2 + 8t - 1$$

Use the parametric derivative formula.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Substitute the two derivatives.

$$\boxed{\frac{dy}{dx} = \frac{6t^2 + 8t - 1}{6t - 3}}$$

Teacher answers & working

Advanced Differentiation · Parametric Differentiation

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S2

For $x(t) = 3t^2 - 2t$ and $y(t) = -4t^3 - 2t^2$, find $\dfrac{d^2y}{dx^2}$.

Answer:

$$\frac{d^2y}{dx^2} = \frac{-72t^2 + 48t + 8}{(6t-2)^3}$$

Working:

Find the first parametric derivative.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{dy}{dx} = \frac{-12t^2 - 4t}{6t - 2}$$

For the second derivative, differentiate dy/dx with respect to t, then divide by dx/dt.

$$\frac{d^2y}{dx^2} = \frac{d}{dt}(dy/dx) \cdot \frac{1}{dx/dt}$$

$$\frac{dx}{dt} = 6t - 2, \quad \frac{dy}{dt} = -12t^2 - 4t$$

$$u = -12t^2 - 4t, \quad v = 6t - 2$$

$$u' = -24t - 4, \quad v' = 6$$

Use the quotient rule.

$$\frac{d}{dt}(u/v) = \frac{u'v - uv'}{v^2}$$

$$\frac{d}{dt}(dy/dx) = \frac{(-24t-4)(6t-2) - (-12t^2-4t)(6)}{(6t-2)^2}$$

Expand the numerator.

$$\frac{d}{dt}(dy/dx) = \frac{-144t^2 + 24t + 8 - (-72t^2 - 24t)}{(6t-2)^2}$$

Collect like terms.

$$\frac{d}{dt}(dy/dx) = \frac{-72t^2 + 48t + 8}{(6t-2)^2}$$

Divide by dx/dt.

$$\frac{d^2y}{dx^2} = \frac{-72t^2 + 48t + 8}{(6t-2)^2} \cdot \frac{1}{6t-2}$$

$$\boxed{\frac{d^2y}{dx^2} = \frac{-72t^2 + 48t + 8}{(6t-2)^3}}$$

Teacher answers & working

Advanced Differentiation · Parametric Differentiation

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S3

Find tangent and normal at $t = 1$ for $x(t) = 4t^2 + 1$ and $y(t) = -3t^3 + 4t + 1$.

Answer:

Tangent: $5x + 8y - 41 = 0$ Normal: $8x - 5y - 30 = 0$

Working:

Substitute t = 1.

$x = 4(1)^2 + (0)(1) + (1)$

$x = 5$

$y = -3(1)^3 + (0)(1)^2 + (4)(1) + (1)$

$y = 2$

Differentiate each parametric equation.

$\frac{dx}{dt} = 8t + (0)$

$\frac{dy}{dt} = -9t^2 + (0)t + (4)$

Substitute the parameter value.

$\frac{dx}{dt} = 8(1) + (0) = 8$

$\frac{dy}{dt} = -9(1)^2 + (0)(1) + (4) = -5$

$m = \frac{dy/dt}{dx/dt}$

$m = \frac{-5}{8} = \frac{-5}{8}$

Take the negative reciprocal.

$m_{normal} = -\frac{8}{-5} = \frac{8}{5}$

Tangent:

$y - y_1 = m(x - x_1)$

$y - (2) = \frac{-5}{8}(x - (5))$

$8(y - (2)) = -5(x - (5))$

$8y - (16) = -5x - (-25)$

$5x + 8y - 41 = 0$

Normal:

$y - y_1 = m(x - x_1)$

$y - (2) = \frac{8}{5}(x - (5))$

$5(y - (2)) = 8(x - (5))$

$5y - (10) = 8x - (40)$

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S3 (continued)

$8x - 5y - 30 = 0$

S4

For $x(t) = -2t^2 + 3t$ and $y(t) = -4t^3 - 2t$, find where tangents are horizontal or vertical.

Answer:

Horizontal: $-12t^2 - 2 = 0$ Vertical: $-4t + 3 = 0$

Working:

For parametric curves, $dy/dx = (dy/dt)/(dx/dt)$.

A horizontal tangent happens when $dy/dt = 0$, provided $dx/dt \neq 0$.

A vertical tangent happens when $dx/dt = 0$, provided $dy/dt \neq 0$.

$$\frac{dy}{dt} = -12t^2 - 2$$

$$\frac{dx}{dt} = -4t + 3$$

$$\boxed{\text{Horizontal: } -12t^2 - 2 = 0 \quad \text{Vertical: } -4t + 3 = 0}$$

E1

Express the area from $t = 1$ to $t = 4$ for $x(t) = -4t^2$, $y(t) = -2t - 4$.

Answer:

$A = \int_1^4 (-2t - 4)(-8t)\,dt$

Working:

Area under a parametric curve is found using $A = \int y\,dx$.

Because x is given in terms of t, replace dx with $\frac{dx}{dt}\,dt$.

$$\frac{dx}{dt} = -8t$$

Substitute $y(t) = -2t - 4$, $dx/dt = -8t$, and the bounds for t.

$$\boxed{A = \int_1^4 (-2t - 4)(-8t)\,dt}$$

Teacher answers & working

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E2

For $x(t) = -2t^2 - 4t$ and $y(t) = -3t^3 - 2t^2 + 4t + 2$, find $\frac{dy}{dx}$.

Answer:

$$\frac{dy}{dx} = \frac{-9t^2 - 4t + 4}{-4t - 4}$$

Working:

For parametric equations, differentiate both x and y with respect to t.

$$\frac{dx}{dt} = -4t - 4$$

$$\frac{dy}{dt} = -9t^2 - 4t + 4$$

Use the parametric derivative formula.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Substitute the two derivatives.

$$\boxed{\frac{dy}{dx} = \frac{-9t^2 - 4t + 4}{-4t - 4}}$$

Teacher answers & working

Advanced Differentiation · Parametric Differentiation

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E3

For $x(t) = -2t^2$ and $y(t) = -3t^3 - 3t^2$, find $\frac{d^2y}{dx^2}$.

Answer:

$$\frac{d^2y}{dx^2} = \frac{36t^2}{(-4t)^3}$$

Working:

Find the first parametric derivative.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{dy}{dx} = \frac{-9t^2 - 6t}{-4t}$$

For the second derivative, differentiate dy/dx with respect to t, then divide by dx/dt.

$$\frac{d^2y}{dx^2} = \frac{d}{dt}(dy/dx) \cdot \frac{1}{dx/dt}$$

$$\frac{dx}{dt} = -4t, \quad \frac{dy}{dt} = -9t^2 - 6t$$

$$u = -9t^2 - 6t, \quad v = -4t$$

$$u' = -18t - 6, \quad v' = -4$$

Use the quotient rule.

$$\frac{d}{dt}(u/v) = \frac{u'v - uv'}{v^2}$$

$$\frac{d}{dt}(dy/dx) = \frac{(-18t - 6)(-4t) - (-9t^2 - 6t)(-4)}{(-4t)^2}$$

Expand the numerator.

$$\frac{d}{dt}(dy/dx) = \frac{72t^2 + 24t - (36t^2 + 24t)}{(-4t)^2}$$

Collect like terms.

$$\frac{d}{dt}(dy/dx) = \frac{36t^2}{(-4t)^2}$$

Divide by dx/dt.

$$\frac{d^2y}{dx^2} = \frac{36t^2}{(-4t)^2} \cdot \frac{1}{-4t}$$

$$\boxed{\frac{d^2y}{dx^2} = \frac{36t^2}{(-4t)^3}}$$